MSG No. 177.150 $P6221'$ [Type II, hexagonal]

Table 1: Wyckoff site: 1a, site symmetry: 6221'

No.	position	mapping
1	[0, 0, 0]	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]

Table 2: Wyckoff site: 1b, site symmetry: 6221'

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]

Table 3: Wyckoff site: 2c, site symmetry: 3.21'

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	[1,3,5,10,11,12,13,15,17,22,23,24]
2	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	[2,4,6,7,8,9,14,16,18,19,20,21]

Table 4: Wyckoff site: 2d, site symmetry: 3.21'

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	[1,3,5,10,11,12,13,15,17,22,23,24]
2	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	[2,4,6,7,8,9,14,16,18,19,20,21]

Table 5: Wyckoff site: 2e, site symmetry: 6..1'

No.	position	mapping
1	$[0, 0, z]$	[1,2,3,4,5,6,13,14,15,16,17,18]
2	$[0, 0, -z]$	[7,8,9,10,11,12,19,20,21,22,23,24]

Table 6: Wyckoff site: 3f, site symmetry: 2221'

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	[1,4,7,11,13,16,19,23]
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[2,5,9,10,14,17,21,22]
3	$\left[0, \frac{1}{2}, 0\right]$	[3,6,8,12,15,18,20,24]

Table 7: Wyckoff site: 3g, site symmetry: 2221'

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[1,4,7,11,13,16,19,23]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[2,5,9,10,14,17,21,22]

3	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,6,8,12,15,18,20,24]`

Table 8: Wyckoff site: `4h`, site symmetry: `3..1'`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1,3,5,13,15,17]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ z\right]$	`[2,4,6,14,16,18]`
3	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[7,8,9,19,20,21]`
4	$\left[\frac{1}{3},\ \frac{2}{3},\ -z\right]$	`[10,11,12,22,23,24]`

Table 9: Wyckoff site: `6i`, site symmetry: `2..1'`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ z\right]$	`[1,4,13,16]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[2,5,14,17]`
3	$\left[0,\ \frac{1}{2},\ z\right]$	`[3,6,15,18]`
4	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[7,11,19,23]`
5	$\left[0,\ \frac{1}{2},\ -z\right]$	`[8,12,20,24]`
6	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[9,10,21,22]`

Table 10: Wyckoff site: `6j`, site symmetry: `.2.1'`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,7,13,19]`
2	$[x,\ x,\ 0]$	`[2,10,14,22]`
3	$[0,\ x,\ 0]$	`[3,8,15,20]`
4	$[-x,\ 0,\ 0]$	`[4,11,16,23]`
5	$[-x,\ -x,\ 0]$	`[5,9,17,21]`
6	$[0,\ -x,\ 0]$	`[6,12,18,24]`

Table 11: Wyckoff site: `6k`, site symmetry: `.2.1'`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[1,7,13,19]`
2	$\left[x,\ x,\ \frac{1}{2}\right]$	`[2,10,14,22]`
3	$\left[0,\ x,\ \frac{1}{2}\right]$	`[3,8,15,20]`
4	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[4,11,16,23]`
5	$\left[-x,\ -x,\ \frac{1}{2}\right]$	`[5,9,17,21]`
6	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[6,12,18,24]`

Table 12: Wyckoff site: 6l, site symmetry: ..21'

No.	position	mapping
1	$[x, -x, 0]$	[1,12,13,24]
2	$[2x, x, 0]$	[2,7,14,19]
3	$[x, 2x, 0]$	[3,10,15,22]
4	$[-x, x, 0]$	[4,8,16,20]
5	$[-2x, -x, 0]$	[5,11,17,23]
6	$[-x, -2x, 0]$	[6,9,18,21]

Table 13: Wyckoff site: 6m, site symmetry: ..21'

No.	position	mapping
1	$\left[x, -x, \frac{1}{2}\right]$	[1,12,13,24]
2	$\left[2x, x, \frac{1}{2}\right]$	[2,7,14,19]
3	$\left[x, 2x, \frac{1}{2}\right]$	[3,10,15,22]
4	$\left[-x, x, \frac{1}{2}\right]$	[4,8,16,20]
5	$\left[-2x, -x, \frac{1}{2}\right]$	[5,11,17,23]
6	$\left[-x, -2x, \frac{1}{2}\right]$	[6,9,18,21]

Table 14: Wyckoff site: 12n, site symmetry: 11'

No.	position	mapping
1	$[x, y, z]$	[1,13]
2	$[x-y, x, z]$	[2,14]
3	$[-y, x-y, z]$	[3,15]
4	$[-x, -y, z]$	[4,16]
5	$[-x+y, -x, z]$	[5,17]
6	$[y, -x+y, z]$	[6,18]
7	$[x-y, -y, -z]$	[7,19]
8	$[y, x, -z]$	[8,20]
9	$[-x, -x+y, -z]$	[9,21]
10	$[x, x-y, -z]$	[10,22]
11	$[-x+y, y, -z]$	[11,23]
12	$[-y, -x, -z]$	[12,24]